

Ezra N. Msolla

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Education

University of Texas at Austin

Aug 2025 –

PhD. in Physics

University of Toronto

Sept 2021 – May 2025

Hon Bsc. in Astronomy and Physics

Thesis: Bayesian emulator optimization for cosmology: Applications to primordial spectra

Advisor: Renée Hložek

Research Experience

Constraints on Parity Violation During Inflation

Toronto, CA

University of Toronto, DAMTP-Cambridge University

Sept 2024 – Present

- Supervised by Dr. Ayngaran Thavanesan
- Developed analytical frameworks to test parity-violating mechanisms in inflationary scenarios.
- Created a model-agnostic constraint method for cosmological observables, laying groundwork for testing with upcoming CMB polarization data.

Fisher Forecasts for the Simons Observatory

Toronto, CA

Department of Astronomy, University of Toronto

Sept 2024 – Sept 2025

- Supervised by Prof. Renée Hložek
- Developing a custom neural network emulator of Einstein-Boltzmann solvers for a fully differentiable analysis pipeline.
- Built an auto-differentiable forecast pipeline for next-generation Cosmic Microwave Background detectors.

Neutrino Self-Interactions in Cosmology

New York City, USA

Flatiron Institute, University of Southern California

May 2024 – Aug 2024

- Supervised by Prof. Vera Gluscevic
- Designed a custom simulation pipeline to study large-scale structure within neutrino self-interaction cosmologies.
- Demonstrated how neutrino interactions reshape the BAO feature, providing insight into reconciling H_0 and S_8 tensions.

Bayesian Analysis in Cosmology

Toronto, CA

Department of Physics, University of Toronto

Sept 2023 – Apr 2024

- Supervised by Dr. Vincent MacKay
- Built a Monte Carlo pipeline for cosmological parameter estimation on Planck 2018 data.
- Produced consistent parameter estimates, and improved science communication skills by presenting findings in colloquia and workshops.

Publications and Talks

J = Journal, C = Conference

- [J.1] **Msolla E.**, Thavanesan A. "Analytical approximations for curved primordial tensor spectra" Nov 2025
([arXiv:2511.10644](https://arxiv.org/abs/2511.10644) )

Honours and Awards

Flatiron Institute Research Fellowship
Simons-NSBP Scholars Program (SNSP)

May 2024

Momentum Builders Scholarship
Physics Department, University of Toronto

September 2023

International Admissions Award
University of Toronto

September 2021

Supervision and mentoring

Physics Directed Reading Program Mentor, UT Austin
Exploring topics in Theoretical Cosmology

September 2025 - Present

Teaching Assistant, UT Austin
PHY105N: Introductory electricity, magnetism, and optics laboratory.

Fall 2025 - Present

Research Mentor: Visions of Science
Supervising a high school senior introductory astronomy project

May 2025 - Aug 2025

IN-GAP: Outreach Supervisor
Facilitating Undergraduate Physics Workshops in East Africa

September 2024 - Present

Miscellaneous

Computing Languages: Python, C++, C, FORTRAN

Programs and Tools: Latex, CLASS, CAMB, TensorFlow, Cudl, CosmoPower, JAX

Certifications: NVIDIA Fundamentals of Deep Learning

Languages: English (Native), Swahili (Native), Italian (Fluent), French (Intermediate), Spanish (Conversational)